# MANUFACTURING ANALYTICS PROJECT

Sales • Production • Inventory • Product Cost Analysis

**Portfolio Project Report**

Dataset period: 1 January 2024 – 30 September 2026

Currency: USD

Company: fictional mid-size paints and coatings manufacturer

Tools: Power Query, SQL, data modeling, DAX, Power BI

Prepared from the supplied project CSVs and data dictionary. Results are estimates from synthetic data and should not be presented as audited company accounts.

# 1. Executive Summary

This project analyzes a fictional paints and coatings manufacturer using sales orders, sales lines, production runs, material usage, purchases, stock counts, products, customers, raw materials, bills of materials, and suppliers. The analysis period is 1 January 2024 through 30 September 2026. Delivered sales total **$10,692,616** across **378,231 units**. Cost and profit figures are estimated from recorded batch costs and standard raw-material costs.

Using product-level estimated production cost per unit, estimated delivered-sales COGS is **$8,165,898**, estimated profit is **$2,526,718**, and estimated profit margin is **23.63%**. These are project estimates, not net profit or audited financial results. The three Power BI report pages reconcile to these figures.

### Key observations

- **Sales:** Interior Emulsion Paint is the highest-revenue product at $2,127,404. Industrial Thinner ranks third in revenue ($1,586,677) but has a negative estimated margin of -0.92% (about -$14,558).
- **Margins:** Wall Primer (14.74%) and Wall Putty (10.13%) are also well below the 23.63% average. Metallic Specialty Paint has the highest margin at 47.64%.
- **Inventory:** five materials show physical stock below expected stock, led by M04 (-2.1%), M09 (-1.5%) and M03 (-1.2%). The other ten materials match within rounding.
- **Cost:** material is about 89% of total production cost, so cost control is mainly a material price question. Cost per unit produced averaged $21.57 and stayed between $21.07 and $22.03 by month.
- **Quality:** the unit-weighted scrap rate is 2.47%.

# 2. Business Problem and Objectives

The project turns operational records into a clear view of sales performance, production activity, inventory reconciliation, and product-level estimated profitability and cost. It answers five core questions:

- How do sales change over time and by product?
- What materials may remain unused, and how do book estimates compare with physical counts?
- Which products have comparatively stronger or weaker estimated margins?
- Do bulk orders receive lower realized prices than regular orders?
- What drives production cost, and is cost per unit stable over time?

# 3. Data and Preparation

The supplied ZIP contains 11 CSV tables plus a data dictionary. The tables describe 1,938 sales-order headers, 1,984 sales-order lines, 590 production runs, 2,653 material-usage records, 653 purchases, 60 customers, 12 finished products, 15 raw materials, 55 bill-of-materials records, 8 suppliers, and 15 physical stock-count records.

### Preparation and modeling

- Standardize text fields (for example, customer types), remove unintended whitespace, and assign correct data types.
- Parse order, production, purchase, signup, and stock-count dates consistently.
- Model order headers to order lines, products to sales lines and production runs, production runs to material usage, and raw materials to purchases, material usage, bills of materials, and stock counts.
- Use a dedicated date table for time-based reporting. The date table is related to the order, production-run and stock-count dates so that the Year slicer filters every page.
- Use delivered orders for revenue and sales-based profit calculations. Cancelled orders are excluded; returned orders are treated explicitly according to the analysis rule.

## 4. SQL Analysis and Key Findings

### 4.1 Delivered sales and yearly trend

| Year | Delivered revenue | Coverage |
|---|---|---|
| 2024 | $3,672,616 | Full year |
| 2025 | $3,979,556 | Full year |
| 2026 | $3,040,443 | Jan–Sep only |

Revenue uses delivered order-line quantity multiplied by the actual unit selling price after discount. The 2026 figure covers only January through September, so it should not be compared with full-year totals without accounting for the shorter period.

### 4.2 Product revenue and estimated margins

| Product | Revenue | Units sold | Est. margin |
|---|---|---|---|
| Interior Emulsion Paint | $2,127,404 | 58,249 | 29.22% |
| Exterior Weatherproof Paint | $1,608,740 | 36,002 | 32.17% |
| Industrial Thinner | $1,586,677 | 79,559 | -0.92% |
| Wall Primer | $1,067,945 | 47,219 | 14.74% |
| Wall Putty | $1,013,089 | 52,093 | 10.13% |
| Waterproofing Compound | $852,492 | 24,157 | 29.82% |

Industrial Thinner generated about $1,586,677 in delivered revenue, but its average realized price of roughly $19.94 per unit is below its estimated cost of $20.13 per unit, giving an estimated loss of about $14,558. This is a signal to investigate, not proof by itself that the product is unprofitable in full accounting terms.

## 4.3 Bulk versus regular orders

A delivered order is classified as Bulk when its total ordered quantity is at least 1,000 units; otherwise it is Regular. In the line-weighted delivered-sales data, average realized price per unit is approximately $29 for Regular orders and $24 for Bulk orders. This is consistent with a lower average realized price on bulk orders, but product mix and estimated contribution should be compared before recommending a discount policy.

## 4.4 Inventory reconciliation

Expected stock is total purchases minus total material usage, relying on the dataset note that opening stock was zero. It is compared with the physical count at 30 September 2026, material by material.

| Material | Physical stock | Expected stock | Gap | Gap % |
|---|---|---|---|---|
| M04 | 74,680 | 76,282 | -1,602 | -2.1% |
| M09 | 2,793 | 2,835 | -43 | -1.5% |
| M03 | 109,712 | 111,044 | -1,333 | -1.2% |
| M12 | 18,802 | 18,954 | -152 | -0.8% |
| M01 | 18,820 | 18,914 | -95 | -0.5% |
| Other 10 materials | match | match | 0 | 0.0% |

**Unit caution:** kilograms and individual items must not be summed into one meaningful quantity. The all-materials total on the dashboard (physical 332,708 vs expected 335,932, a gap of -3,224 or -1.0%) adds quantities across materials and is shown only as an indicator. The material-level gaps above are the real finding. Differences can reflect timing, scrap, counting, recording, or other adjustment issues.

## 4.5 Production cost structure

Total production cost (material + batch labor + batch overhead) is **$8,685,581** for 402,738 units produced, or **$21.57 per unit**. Material makes up **89.0%** of that cost; labor and overhead share the remaining 11%. The unit-weighted scrap rate is **2.47%**. Monthly cost per unit moved in a narrow band, from $21.07 to $22.03, peaking in May 2025.

Total production cost ($8.69M) is higher than estimated COGS ($8.17M) because production cost covers everything made (402,738 units), while COGS covers only what was delivered (378,231 units).

| Product | Cost per unit | Est. margin |
|---|---|---|
| Epoxy Floor Coating | $45.74 | 41.57% |
| Anti-Corrosion Paint | $30.70 | 34.65% |
| Exterior Weatherproof Paint | $30.31 | 32.17% |
| Interior Emulsion Paint | $25.85 | 29.22% |
| Waterproofing Compound | $24.77 | 29.82% |
| Texture Coating | $21.20 | 31.35% |
| Industrial Thinner | $20.13 | -0.92% |
| Wall Primer | $19.28 | 14.74% |
| Wall Putty | $17.48 | 10.13% |
| Wood Varnish | $15.04 | 36.92% |
| Metallic Specialty Paint | $12.28 | 47.64% |
| Enamel Paint | $7.58 | 32.37% |

A high cost per unit does not mean low profit: Epoxy Floor Coating is the most expensive product to make and still has a 41.57% margin. Margin depends on the selling price relative to cost, which is why Industrial Thinner, with a mid-range unit cost, is the one that loses money.

## 5. Power BI Dashboard Overview

The project includes three Power BI report pages, built to give different audiences a clear view of performance. Each page has a Year slicer and a headline or title that updates with the filters, so the written message stays correct as the data changes.

### Page 1 — Executive Overview

**Purpose:** summarize overall sales and profitability. KPI cards show total revenue ($10,692,616), estimated profit ($2,526,718), estimated margin (23.63%) and delivered units (378,231). A table of products is colored by margin, and a dynamic headline names the loss-making product. Charts show the top five products by revenue and the monthly revenue trend, which peaked in November 2025.

### Page 2 — Production & Inventory

**Purpose:** show production activity and inventory position. KPI cards show total production (402,738 units), production runs (590), total material used (6,343,693) and the stock gap. A table compares physical and expected stock by material with a dynamic headline, and a chart ranks materials by gap %. A product chart shows production by product, and the monthly trend peaked in October 2025 at 13,854 units. A recommendation box states the follow-up action.

### Page 3 — Product & Cost Analysis

**Purpose:** explain what drives production cost. KPI cards show total production cost ($8,685,581), cost per unit produced ($21.57), material cost % (89.0%) and scrap % (2.47%). A monthly trend of cost per unit has a dynamic title stating the range and peak month. Two product charts show cost per unit and the cost breakdown into material, labor and overhead. A recommendation box points to material purchase prices.

## 6. Costing Method and Validation Notes

Estimated material cost = material quantity used × standard material cost. Estimated labor and overhead costs are summed from production-run records because the data dictionary states that these are total costs for the whole batch, not per-unit costs. For each product, estimated total production cost is allocated across produced units to derive an estimated cost per unit. Estimated COGS is then delivered units sold × the product-level estimated cost per unit.

| Metric | Source-data estimate |
|---|---|
| Delivered revenue | $10,692,616 |
| Estimated COGS | $8,165,898 |
| Estimated profit | $2,526,718 |
| Estimated profit margin | 23.63% |
| Total production cost (all units made) | $8,685,581 |
| Cost per unit produced | $21.57 |

### Reconciliation of the DAX measures

During the build, the report's profit briefly showed $3,067,180 (28.69%) instead of $2,526,718. The cause was a measure named *Total labor* that was already a per-unit figure (labor cost ÷ units produced), which a second measure then divided by units produced again. Labor therefore dropped out of cost per unit and understated COGS. The per-unit labor measure now divides the batch-total labor cost by units produced once, and all three report pages reconcile to the source-data estimates above.

**Lesson:** name measures by their unit and grain (for example, *Labor Cost Per Unit* rather than *Total labor*), and keep batch totals separate from per-unit values.

**Important limitation:** this is not net profit. The dataset may not include all business costs such as freight, utilities, salaries outside recorded batch labor, taxes, sales-returns treatment, financing, or other operating expenses. Product-level unit costs are estimates and should be reviewed against the business's costing policy and production/sales timing.

## 7. Business Recommendations

1. **Review Industrial Thinner pricing and costs.** Its average realized price (about $19.94) is below its estimated cost per unit ($20.13). Check discounts, material consumption, scrap, and the allocation period before changing prices.

2. **Review Wall Primer and Wall Putty.** Both are in the top five by revenue but have margins of 14.74% and 10.13%, well below the 23.63% average. Together with Industrial Thinner, these three products bring in about $3.7 million in revenue.
3. **Focus cost control on material purchase prices.** Material is 89% of production cost, and cost per unit has been stable at $21.07 to $22.03, so savings will come mainly from supplier pricing. Start with the highest-cost products: Epoxy Floor Coating ($45.74 per unit), Anti-Corrosion Paint ($30.70) and Exterior Weatherproof Paint ($30.31).
4. **Reconcile inventory at material level.** Recount and review usage records for M04, M09 and M03, which show the largest gaps (-2.1%, -1.5%, -1.2%). Keep units separate and document adjustments.
5. **Evaluate bulk discounts using contribution, not price alone.** Compare realized prices, product mix, cost per unit, and order size before adjusting discount thresholds.
6. **Improve cost reporting governance.** Keep batch-total labor and overhead separate from per-unit costs, and make the unit and grain explicit in measure names and documentation.
7. **Add regular refresh and exception checks.** Track unusual negative margins, unexpectedly large stock gaps, missing relationships, and incomplete periods.

## 8. Limitations and Assumptions

- The supplied dataset is synthetic and represents a fictional manufacturer.
- The period ends on 30 September 2026; the 2026 revenue total is partial-year.
- The inventory formula assumes opening stock of zero and may not account for every real-world adjustment.
- Inventory quantities are material- and unit-specific. The all-materials stock total mixes units and is indicative only.
- Standard material costs are used as an estimate; actual purchase prices may vary.
- Labor and overhead values are batch totals per the data dictionary.
- Estimated profit excludes any costs not represented in the supplied data.
- Production cost covers all units made, while COGS covers delivered units only, so the two totals differ by design.

## 9. Conclusion

This project demonstrates an end-to-end analytics workflow across data preparation, relational modeling, SQL business analysis, DAX measures, and interactive Power BI reporting. It gives management a basis to investigate sales trends, product-level margins, bulk pricing, inventory differences and the drivers of production cost. The cost measures have been reconciled to the source data, and all estimates are labeled as estimates from synthetic data.

## Appendix — Main measures in business logic

- **Delivered revenue:** sum of delivered quantity × actual unit selling price.
- **Estimated material cost:** sum of material usage quantity × standard material cost.
- **Estimated batch labor cost:** sum of recorded labor cost per production run.
- **Estimated batch overhead cost:** sum of recorded overhead cost per production run.
- **Estimated total production cost:** material + batch labor + batch overhead.
- **Cost per unit produced:** total production cost ÷ units produced.
- **Material cost %:** material cost ÷ total production cost.
- **Scrap %:** sum of (units produced × run scrap rate) ÷ units produced.
- **Estimated COGS:** delivered units sold × product-level estimated cost per unit.
- **Estimated profit:** delivered revenue – estimated COGS.
- **Estimated margin:** estimated profit ÷ delivered revenue × 100.
- **Stock gap and gap %:** physical stock – expected stock, and the gap ÷ expected stock, by material.